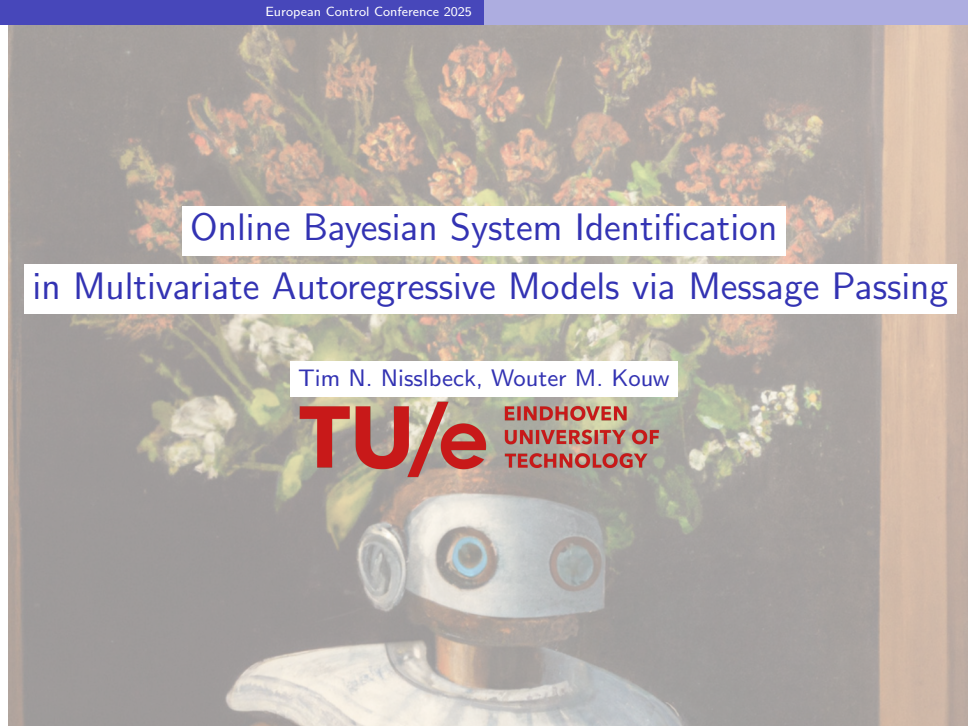


The background of the slide is a painting of a bouquet of flowers, including red and white blossoms, with green leaves. In the foreground, at the bottom center, is a small, stylized robot with a metallic head, large blue eyes, and a silver body.

Online Bayesian System Identification in Multivariate Autoregressive Models via Message Passing

Tim N. Nisslbeck, Wouter M. Kouw

TU/e

**EINDHOVEN
UNIVERSITY OF
TECHNOLOGY**

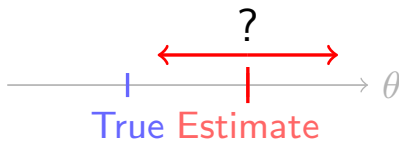
Motivation

Classical ID $\rightarrow$ point estimates



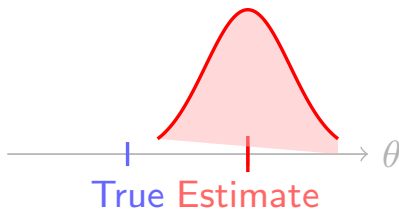
Motivation

Lacks uncertainty quantification



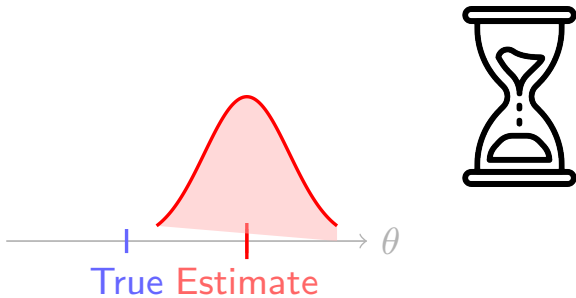
Motivation

Bayesian ID $\rightarrow$ full posteriors

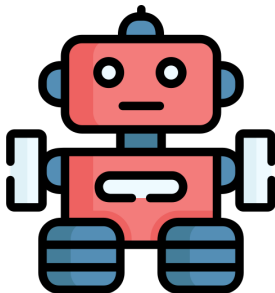


Motivation

Most Bayes methods are offline

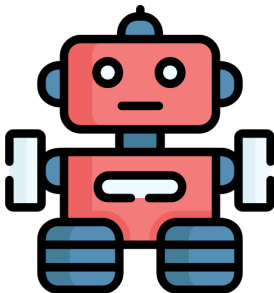


Motivation



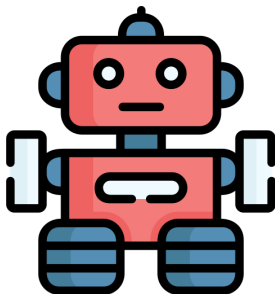
Real-time (Bayesian) Inference?

Motivation



Message Passing: (event-based) updates

Motivation

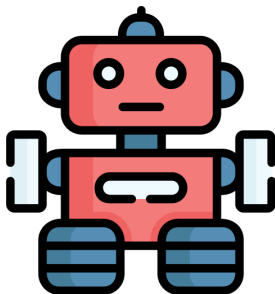


Trigger update if:



Message Passing: (event-based) updates

Motivation

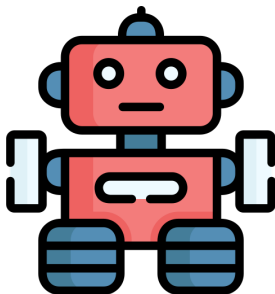


Trigger update if:

- ✓ Resources available
-

Message Passing: (event-based) updates

Motivation



Trigger update if:

- ✓ Resources available
- ✓ Uncertainty high

Message Passing: **Local** (event-based) updates

Problem

Dynamical systems

We study discrete-time state-space systems defined by:

- **State transition:** $z_t = f(z_{t-1}, u_t)$

where at time t

- $z_t \in \mathbb{R}^{D_z}$ state
- $u_t \in \mathbb{R}^{D_u}$ control signal

Problem

Dynamical systems

We study discrete-time state-space systems defined by:

- **State transition:** $z_t = f(z_{t-1}, u_t)$
- **Measurement:** $y_t = g(z_t) + e_t$

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Goal

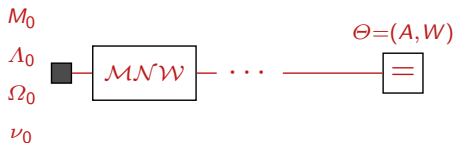
Predict future outputs y_τ for $\tau > t$ given future inputs u_τ using a model of the system dynamics.

Inference as Message Passing on Factor Graphs

$$y_t = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$

Inference as Message Passing on Factor Graphs

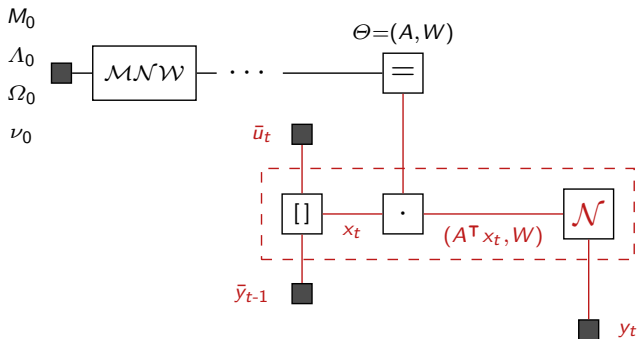
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Prior

Inference as Message Passing on Factor Graphs

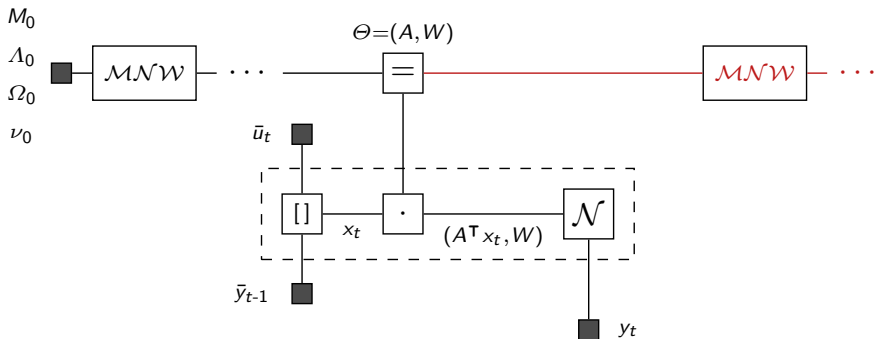
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Likelihood

Inference as Message Passing on Factor Graphs

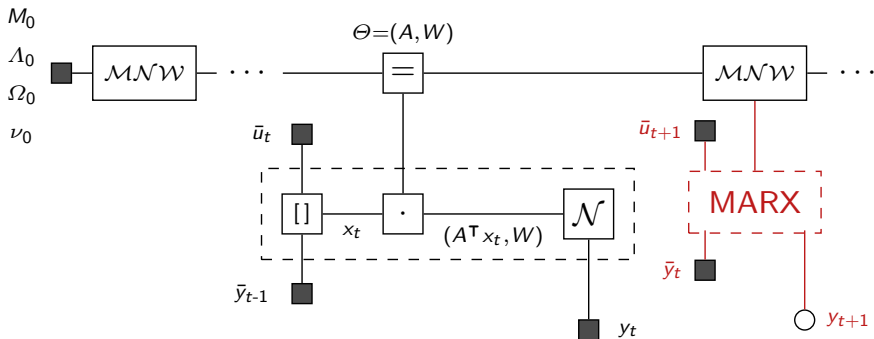
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Posterior

Inference as Message Passing on Factor Graphs

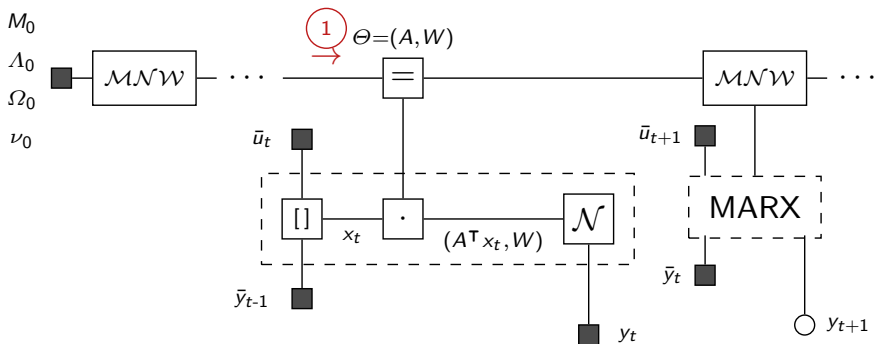
$$y_t = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$



Posterior Predictive

Inference as Message Passing on Factor Graphs

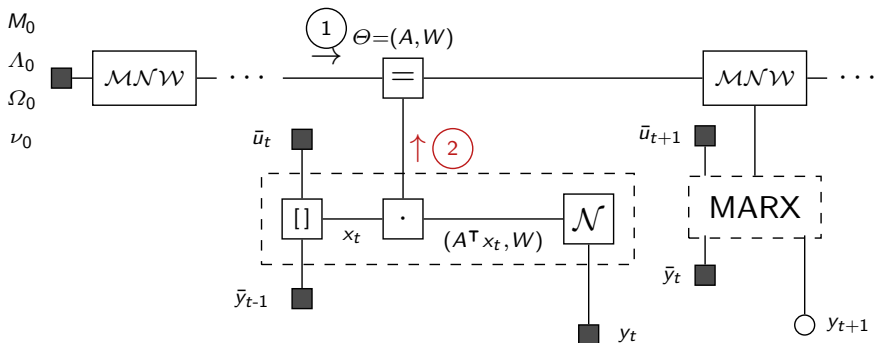
$$y_t = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$



Prior message

Inference as Message Passing on Factor Graphs

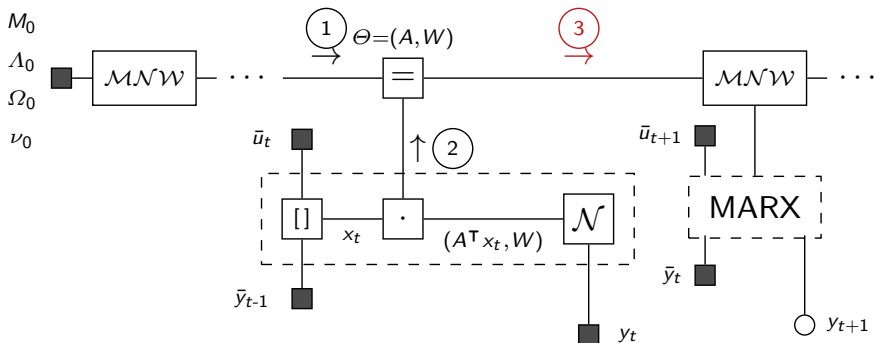
$$y_t = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$



Likelihood message

Inference as Message Passing on Factor Graphs

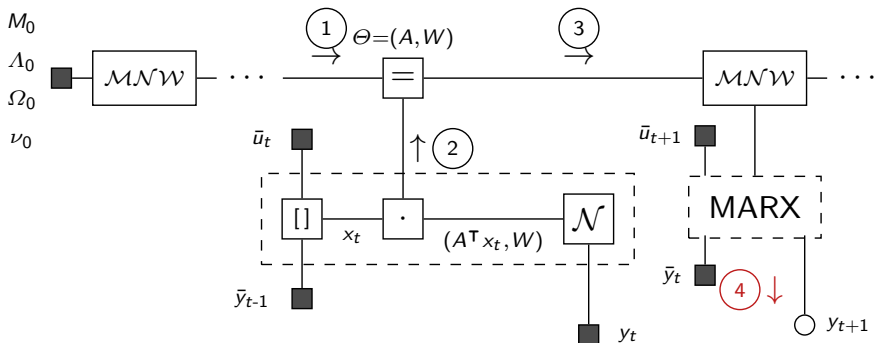
$$y_t = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$



Posterior message

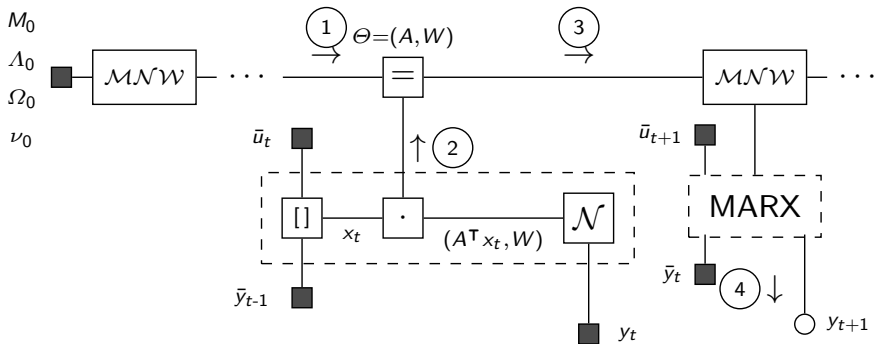
Inference as Message Passing on Factor Graphs

$$y_t = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$



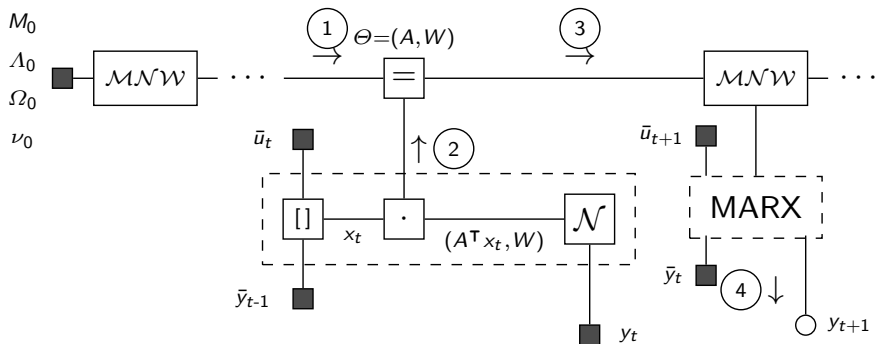
Posterior predictive message

$$y_t = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$



Exact **Recursive** Inference as Message Passing on Factor Graphs

$$y_t = A^T x_t + e_t \quad e_t \sim \mathcal{N}(0, W^{-1})$$



Experiments

- Verification: Synthetic MARX system with **known** parameters

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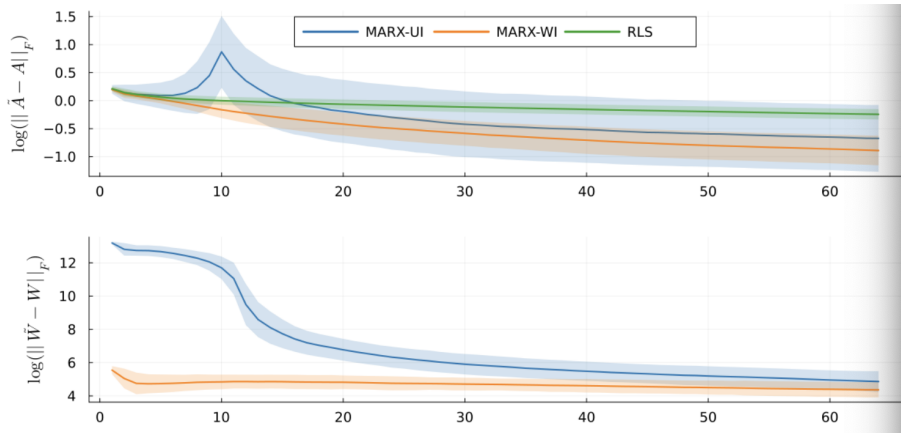
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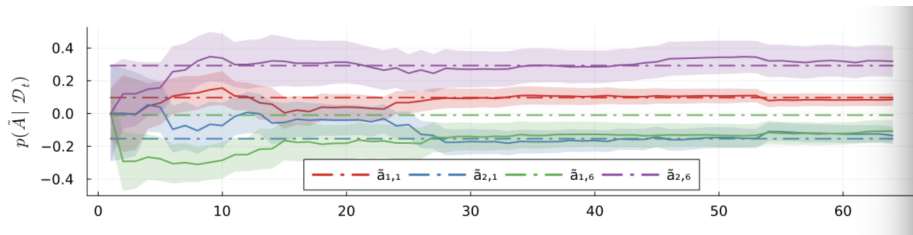
- Validation: Synthetic double mass-spring-damper system with unknown parameters
- Baseline: Recursive Least-Squares (RLS)
- Use of two priors: uninformative (MARX-UI) and weakly informative (MARX-WI)

	M_0	Λ_0	Ω_0	ν_0
Uninformative	$0_{D_x \times D_y}$	$1 \times 10^{-4} \cdot I_{D_x}$	$1 \times 10^{-5} \cdot I_{D_y}$	$D_y + 3$
Weakly informative	$0_{D_x \times D_y}$	$1 \times 10^{-1} \cdot I_{D_x}$	$1 \times 10^{-2} \cdot I_{D_y}$	$D_y + 3$

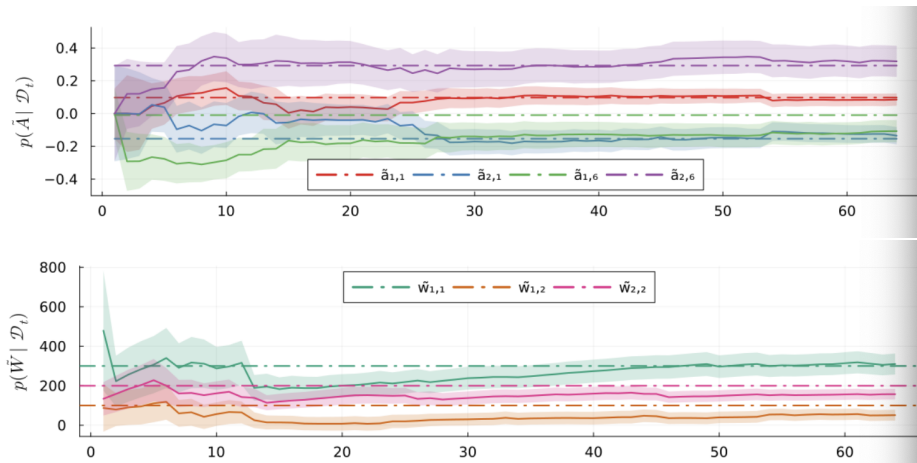
Results



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Discussion & Future Work

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 - interpretability
 - extensibility (e.g. hierarchical or time-varying models)

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- Future work: distributed inference: multi-agent, federated identification

<https://github.com/biaslab/ECC2025-MARXEFE>



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Thank you. Questions?